

The first-passage time density for the diffusion model with variable drift and variable starting point

Kiante Fernandez*

Department of Psychology, University of California, Los Angeles, United States

Abstract

The Ratcliff diffusion model is now arguably the most widely applied model for response time data. Its major advantage is its description of both response times and the probabilities for correct as well as incorrect responses. The model assumes a Wiener process with drift between two constant absorbing barriers. The first-passage times at the upper and lower boundary describe the responses in simple two-choice decision tasks, for example, in experiments with perceptual discrimination or memory search. In applications of the model, the drift rate and the starting point of the Wiener process are usually assumed to vary across trials, normally and uniformly respectively. This extra flexibility allows accounting for the slow and fast errors that often occur in response time experiments. The drift integral was solved analytically by Blurton, Kesselmeier and Gondan (2017); the starting-point integral has so far been evaluated numerically in every implementation, and Tuerlinckx (2004) concluded that no closed form exists. Here we show that this conclusion holds only for the large-time series representation. In the small-time representation the starting-point integral is an ordinary Gaussian moment, and we present a closed-form expression for the first-passage time density with normally distributed drift and uniformly distributed starting point, together with a large-time counterpart in terms of the Faddeeva function that is stable where the small-time series is not. The result is verified symbolically, to 48 digits against direct double integration, against two independent reference implementations, against process simulation, and is machine-checked in Lean 4. Every term is a combination of exponentials, the normal distribution

*Corresponding author.

Email address: `kiantef@ucla.edu` (Kiante Fernandez)

function, and the Faddeeva function, so analytic gradients with respect to all parameters exist. The full seven-parameter likelihood then requires a single one-dimensional quadrature over the non-decision-time window in place of two-dimensional cubature.

Keywords: Diffusion model, Response time modeling, First-passage time, Starting point variability

1. Background

The diffusion model for response times (Ratcliff, 1978) describes a two-choice decision as a Wiener process with drift v and unit diffusion coefficient¹ evolving between two absorbing barriers at 0 and $a > 0$ from a starting point aw , $0 < w < 1$. The first-passage times at the two barriers are the predicted response times for the two alternatives. To account for the fast and slow errors observed in most response time experiments, the model is applied with across-trial variability in three of its quantities (Ratcliff, 1978; Ratcliff and Rouder, 1998; Ratcliff and Tuerlinckx, 2002): the drift rate is normally distributed, $v \sim N(\nu, \eta^2)$; the starting point is uniformly distributed, $w \sim U(w_1, w_2)$, with Ratcliff's $s_z = a(w_2 - w_1)$; and the non-decision time is uniformly distributed over a window of width s_t . The predicted density of a response time is then a triple integral of the constant-parameter first-passage time density over these three distributions.

Of the three integrals, only the drift integral has been available in closed form. The first-passage time density with normally distributed drift is given by Horrocks and Thompson (2004) and, for the lower barrier, reads (Blurton et al., 2017, Eq. 1)

$$g(t \mid \nu, \eta^2, a, w) = \frac{1}{\sqrt{2\pi t^3 S}} \exp\left[\frac{-\nu^2 t - 2\nu a w + \eta^2 (a w)^2}{2S}\right] \times \sum_{j=0}^{\infty} (-1)^j r_j \exp\left(-\frac{r_j^2}{2t}\right), \quad (1)$$

where $S := 1 + \eta^2 t$, $r_j = ja + aw$ for even j and $r_j = ja + a(1 - w)$ for odd j . Blurton et al. (2017) derived the corresponding cumulative distribution

¹For diffusion coefficient $\sigma^2 \neq 1$ all results below hold after the substitution $(v, \eta^2, a) \rightarrow (v/\sigma, \eta^2/\sigma^2, a/\sigma)$ (Blurton et al., 2017).

function. Both results are stated in the small-time representation of the first-passage time series (Navarro and Fuss, 2009; Gondan et al., 2014), that is, as a sum over image charges r_j with Gaussian factors $\exp(-r_j^2/2t)$.

The starting-point and non-decision-time integrals have remained numerical. Tuerlinckx (2004, pp. 706–708) worked in the large-time representation of Ratcliff (1978), in which the starting point enters through $\sin(k\pi w)$, integrated the starting point analytically at fixed drift, and concluded that the drift integral then has to be approximated by quadrature. Every current implementation of the seven-parameter model integrates the starting point numerically: DMAT (Vandekerckhove and Tuerlinckx, 2008), fast-dm and rtdists (Voss and Voss, 2007; Singmann et al., 2022), HDDM (Wiecki et al., 2013), WienR (Hartmann and Klauer, 2021), and the Stan implementation of Henrich et al. (2024). Blurton et al. (2017) noted that adding starting-point and non-decision-time variability to their result “requires numerical evaluation of two integrals” (p. 11).

In this note we show that the starting-point integral of the density (1) has a closed form. The reason it was missed is representational. In the large-time series, after the drift has been integrated, the starting point appears in a Gaussian factor with a *positive* coefficient of w^2 and the integral against $\sin(k\pi w)$ lands on the imaginary error function; this is the obstruction Tuerlinckx encountered. In the small-time series the factor $\exp(-r_j^2/2t)$ contributes a negative coefficient of w^2 that always dominates the positive one produced by the drift integral, so the integrand is a linear function times a decaying Gaussian, and the integral is a Gaussian moment with a closed form. Section 2 gives the result and its derivation, Section 3 the large-time counterpart that is needed for numerical stability at large t , Section 4 the truncation rule, Section 5 the partial derivatives, Section 6 the resulting seven-parameter likelihood, and Section 7 the numerical verification and a comparison of computational cost with existing software.

2. The density with variable drift and variable starting point

Let the starting point be uniformly distributed on $[w_1, w_2]$, $0 \leq w_1 < w_2 \leq 1$. The lower-barrier first-passage time density of the resulting process is

$$g(t \mid \nu, \eta^2, a, w_1, w_2) := \frac{1}{w_2 - w_1} \int_{w_1}^{w_2} g(t \mid \nu, \eta^2, a, w) \, dw. \quad (2)$$

Because $r_j \geq ja$ for $w \in [0, 1]$, the j -th term of (1) is bounded in absolute value by $(j+1)a \exp(-j^2 a^2/2t)$ times a constant independent of w and j (Appendix B), so the series converges absolutely and uniformly on $[w_1, w_2]$ and summation and integration may be exchanged. It therefore suffices to integrate one term.

2.1. The j -th term is a Gaussian moment

Write $r_j = p_j + q_j w$ with $p_j = ja$, $q_j = a$ for even j , and $p_j = (j+1)a$, $q_j = -a$ for odd j . Collecting the w -dependence of the j -th term of (1), that is, of $\exp[(-2\nu a w + \eta^2 a^2 w^2)/2S] r_j \exp(-r_j^2/2t)$, gives a quadratic exponent in w ,

$$(p_j + q_j w) \exp\left(-\frac{1}{2} A w^2 + B_j w + C_j\right), \quad (3)$$

with

$$A = \frac{a^2}{tS}, \quad B_j = -\frac{\nu a}{S} \mp \frac{k_j a^2}{t}, \quad C_j = -\frac{\nu^2 t}{2S} - \frac{k_j^2 a^2}{2t}, \quad (4)$$

where $k_j = j$ and the upper sign apply for even j , and $k_j = j+1$ and the lower sign apply for odd j . The coefficient of w^2 deserves attention. The drift integral contributes $+\eta^2 a^2/2S$, a growing Gaussian in w , whereas the image-charge factor contributes $-a^2/2t$, and

$$\frac{a^2}{2t} - \frac{\eta^2 a^2}{2S} = \frac{a^2(S - \eta^2 t)}{2tS} = \frac{a^2}{2tS} > 0 \quad (5)$$

for every $t > 0$, every η , and every j . The combined exponent is therefore a decaying Gaussian with $A > 0$. This is the whole content of the result: in the small-time representation the starting-point integrand is a linear function times a proper Gaussian, and such integrals have closed forms in terms of $\exp$ and Φ .

2.2. The result

For $A > 0$ and $m := B/A$,

$$\begin{aligned} & \int_{w_1}^{w_2} (p + qw) e^{-\frac{1}{2} A w^2 + B w + C} dw \\ &= (p + qm) \sqrt{\frac{2\pi}{A}} e^{C+B^2/2A} \left[\Phi(\sqrt{A}(w_2 - m)) - \Phi(\sqrt{A}(w_1 - m)) \right] \\ & \quad - \frac{q}{A} \left[e^{C+Bw_2 - \frac{1}{2} A w_2^2} - e^{C+Bw_1 - \frac{1}{2} A w_1^2} \right], \end{aligned} \quad (6)$$

76 where Φ denotes the standard normal distribution function (Appendix A).
 77 Applying (6) term by term with the coefficients (4) gives the lower-barrier
 78 first-passage time density with normally distributed drift and uniformly dis-
 79 tributed starting point,

$$g(t | \nu, \eta^2, a, w_1, w_2) = \frac{1}{(w_2 - w_1)\sqrt{2\pi t^3 S}} \sum_{j=0}^{\infty} (-1)^j I_j, \quad (7)$$

$$I_j = \int_{w_1}^{w_2} (p_j + q_j w) e^{-\frac{1}{2}Aw^2 + B_j w + C_j} dw,$$

80 with I_j given in closed form by (6). The density at the upper barrier is
 81 $g(t | -\nu, \eta^2, a, 1 - w_2, 1 - w_1)$. For $w_2 - w_1 \rightarrow 0$ the result reduces to (1),
 82 and for $\eta = 0$ it reduces to the starting-point integral of the constant-drift
 83 density, so it can be used in a fitting routine regardless of whether either
 84 form of variability is present in the data.

85 Two remarks on evaluation. First, for large j the centre $m = B_j/A$ of
 86 the Gaussian in (6) lies far outside $[w_1, w_2]$, so that $e^{C+B^2/2A}$ is very large
 87 and the bracketed difference of Φ values is very small. The product must be
 88 formed in logarithmic space, using $\log \Phi$ and a tail-aware difference; with that
 89 precaution the expression is stable for all parameter values we have tested.
 90 Second, the identity $\sqrt{2\pi} e^{C+B^2/2A} \phi(\sqrt{A}(w_i - m)) = e^{C+Bw_i - \frac{1}{2}Aw_i^2}$, where ϕ
 91 is the standard normal density, shows that the two brackets in (6) share their
 92 exponential factors; this is used in the derivatives of Section 5.

93 3. Large-time representation

94 Like every small-time series, (7) loses precision through cancellation when
 95 t/a^2 is large; in double precision the loss becomes visible for $t/a^2 \gtrsim 2$.
 96 A large-time counterpart is obtained from the large-time series of Ratcliff
 97 (1978). Integrating $\exp(-vaw - v^2 t/2)$ against the $N(\nu, \eta^2)$ drift density
 98 produces exactly the prefactor of (1), so the large-time density with nor-
 99 mally distributed drift is

$$g(t | \nu, \eta^2, a, w) = \frac{\pi}{a^2 \sqrt{S}} \exp \left[\frac{-\nu^2 t - 2\nu a w + \eta^2 (aw)^2}{2S} \right]$$

$$\times \sum_{k=1}^{\infty} k e^{-k^2 \pi^2 t / 2a^2} \sin(k\pi w). \quad (8)$$

Here there is no image-charge factor to compensate the growing Gaussian, and the starting-point integral of the k -th term is $\int_{w_1}^{w_2} e^{\kappa w^2 + \lambda w} \sin(k\pi w) dw$ with $\kappa = \eta^2 a^2 / 2S > 0$ and $\lambda = -\nu a / S$. Written as the imaginary part of $\int e^{\kappa w^2 + \mu_k w} dw$ with $\mu_k = \lambda + ik\pi$, the antiderivative is an imaginary error function of complex argument, which overflows. It is, however, bounded when expressed through the Faddeeva function $w(z) = e^{-z^2} \operatorname{erfc}(-iz)$ (Abramowitz and Stegun, 1964, 7.1.3):

$$\int_{w_1}^{w_2} e^{\kappa w^2 + \mu_k w} dw = \frac{-i\sqrt{\pi}}{2\sqrt{\kappa}} \left[e^{\kappa w^2 + \mu_k w} w(z_k(w)) \right]_{w_1}^{w_2}, \quad z_k(w) = \sqrt{\kappa} w + \frac{\mu_k}{2\sqrt{\kappa}}. \quad (9)$$

Since $\operatorname{Im} z_k = k\pi / 2\sqrt{\kappa} > 0$, $w(z_k)$ lies in the upper half plane where it is bounded by one, and the product with $e^{\kappa w^2 + \mu_k w}$ is of the size of the integrand. For $\eta = 0$ the bracket reduces to $(e^{\mu_k w_2} - e^{\mu_k w_1}) / \mu_k$. The large-time density with uniformly distributed starting point is then

$$g(t \mid \nu, \eta^2, a, w_1, w_2) = \frac{\pi e^{-\nu^2 t / 2S}}{a^2 \sqrt{S} (w_2 - w_1)} \sum_{k=1}^{\infty} k e^{-k^2 \pi^2 t / 2a^2} \operatorname{Im} \int_{w_1}^{w_2} e^{\kappa w^2 + \mu_k w} dw. \quad (10)$$

Fast and accurate algorithms for $w(z)$ are standard (Zaghloul and Ali, 2011) and available in scientific libraries. In our implementation the two representations are switched at $t/a^2 = 1$; they agree to 1.5×10^{-11} relative across the switch. We stress that (9) must be used in the Faddeeva form: written with the imaginary error function it loses of the order of $\pi^2 k^2 / 4\kappa$ nats to cancellation and returns meaningless values when the series is truncated at a fixed large K .

4. Convergence

The j -th term of (7) is bounded in absolute value by $c(j+1)a \exp(-j^2 a^2 / 2t)$, and the k -th term of the large-time series (10) by $c' k \exp(-k^2 \pi^2 t / 2a^2)$, where the constants c and c' do not depend on the index (Appendix B). Both bounds decrease super-exponentially, so the truncation error is of the order of the first neglected term, and for a tolerance ε the series may be cut at

$$J = \left\lceil \frac{\sqrt{2t \log(1/\varepsilon)}}{a} \right\rceil + 1, \quad K = \left\lceil a \sqrt{\frac{2 \log(1/\varepsilon)}{\pi^2 t}} \right\rceil + 1. \quad (11)$$

125 With $\varepsilon = 10^{-12}$ the small-time series needs $J \leq 9$ terms for $t/a^2 \leq 1$, and
 126 the large-time series needs $K \leq 4$ terms for $t/a^2 \geq 1$. Evaluating (7) with
 127 $\varepsilon = 10^{-12}$ and with $\varepsilon = 10^{-40}$ on the 400 parameter sets of Section 7 gives
 128 results that differ by at most 2.2×10^{-16} .

129 5. Partial derivatives

130 Gradient information is required by Hamiltonian Monte Carlo and by
 131 gradient-based optimisers, and Hartmann and Klauer (2021) derived it for
 132 the constant-parameter model. Every term of (7) and (10) is a composition
 133 of $\exp$, Φ , and w , so the partial derivatives with respect to ν , η , a , w_1 , and w_2
 134 are again closed form. For the small-time representation it is convenient to
 135 differentiate I_j with respect to its seven coefficients and to apply the chain
 136 rule. Writing $G := \sqrt{2\pi/A} e^{C+B^2/2A} [\Phi(\sqrt{A}(w_2 - m)) - \Phi(\sqrt{A}(w_1 - m))]$
 137 and $E_i := e^{C+Bw_i-\frac{1}{2}Aw_i^2}$, so that $I = (p + qm)G - (q/A)(E_2 - E_1)$, the total
 138 differential of I under $(dA, dB, dC, dp, dq, dw_1, dw_2)$ is

$$\begin{aligned} dI &= (dp + m dq + q dm) G + (p + qm) dG \\ &\quad - \left(\frac{dq}{A} - \frac{q dA}{A^2} \right) (E_2 - E_1) - \frac{q}{A} (dE_2 - dE_1), \\ dm &= \frac{dB - m dA}{A}, \quad dK = dC + m dB - \frac{1}{2}m^2 dA, \\ dG &= G \left(dK - \frac{dA}{2A} \right) + E_2 \left[(dw_2 - dm) + \frac{(w_2 - m) dA}{2A} \right] \\ &\quad - E_1 \left[(dw_1 - dm) + \frac{(w_1 - m) dA}{2A} \right], \\ dE_i &= E_i \left[dC + w_i dB - \frac{1}{2}w_i^2 dA + (B - Aw_i) dw_i \right], \end{aligned} \quad (12)$$

139 in which the identity noted at the end of Section 2 has been used to express
 140 dG through E_1 and E_2 rather than through ϕ . The coefficient differentials
 141 follow from (4); for instance $\partial B_j / \partial \nu = -a/S$ and $\partial C_j / \partial \nu = -\nu t/S$, and
 142 A , p_j , q_j do not depend on ν . For the large-time representation only two
 143 identities are needed, both immediate from $\frac{d}{dw} e^{\kappa w^2 + \mu w} = (2\kappa w + \mu) e^{\kappa w^2 + \mu w}$:
 144 with $\mathcal{I} := \int_{w_1}^{w_2} e^{\kappa w^2 + \mu w} dw$ and $\mathcal{E}_i := e^{\kappa w_i^2 + \mu w_i}$,

$$\frac{\partial \mathcal{I}}{\partial \mu} = \frac{(\mathcal{E}_2 - \mathcal{E}_1) - \mu \mathcal{I}}{2\kappa}, \quad \frac{\partial \mathcal{I}}{\partial \kappa} = \frac{(w_2 \mathcal{E}_2 - w_1 \mathcal{E}_1) - \mathcal{I} - \mu \partial \mathcal{I} / \partial \mu}{2\kappa}. \quad (13)$$

145 The derivatives with respect to w_1 and w_2 are, by the fundamental theorem
 146 of calculus, the integrand evaluated at the endpoints, corrected for the fac-
 147 tor $1/(w_2 - w_1)$. All of (12) and (13), and every chain-rule coefficient, are
 148 verified as exact symbolic identities in the supplementary code; the resulting
 149 gradients agree with those of Hartmann and Klauer (2021) as implemented
 150 in WienR to 3×10^{-13} absolute on the parameter grid of Section 7.

151 6. The seven-parameter likelihood

152 With the non-decision time uniformly distributed on $[t_0, t_0 + s_t]$, the den-
 153 sity of an observed response time is

$$f(t) = \frac{1}{s_t} \int_{t-t_0-s_t}^{t-t_0} g(u \mid \nu, \eta^2, a, w_1, w_2) du, \quad (14)$$

154 where the lower limit is replaced by 0 if negative. This integral does not
 155 have a closed form: the non-decision time shifts the time argument, which
 156 enters every term of (7) through $\sqrt{t}$ and $1/t$, and we have not found a
 157 representation in which the shift integrates. What (7) achieves is to reduce
 158 the seven-parameter likelihood from a two-dimensional cubature over (w, u)
 159 to a one-dimensional quadrature over u of a smooth closed-form integrand
 160 on a short interval. Because $g(u)$ rises steeply from zero at $u = 0$ and
 161 the window may include that leading edge, a plain Gauss–Legendre rule
 162 converges slowly; the substitution $u = u_{lo} + (u_{hi} - u_{lo}) s^3$, which clusters
 163 nodes at the lower limit, fixes this. Table 1 gives the maximum relative error
 164 of the rule against adaptive quadrature over the 200 seven-parameter sets
 165 described below. A 32-node rule with the cubic map reaches 1.5×10^{-9} at
 166 a fixed cost of 32 evaluations of (7). The gradient of (14) with respect to
 167 the five parameters of g is the same rule applied to the gradient of g , and
 168 $\partial f / \partial t_0 = [g(u_{lo}) - g(u_{hi})] / s_t$ and $\partial f / \partial s_t = [g(u_{lo}) - f] / s_t$ follow from the
 169 endpoints.

170 7. Verification and efficiency

171 7.1. Verification

172 We checked the result in five independent ways. (i) Every algebraic step,
 173 in particular the antiderivative (6), the inequality (5), the coefficient ta-
 174 ble (4), the drift integral leading to (1), and all derivative formulae of Sec-
 175 tion 5, is verified as an exact symbolic identity in a computer algebra system,

Table 1: Maximum relative error of Gauss–Legendre quadrature for (14) against adaptive quadrature, 200 random parameter sets, densities above 10^{-3} .

Nodes	Plain	With cubic map
8	4.1×10^{-3}	4.3×10^{-3}
16	7.3×10^{-5}	3.5×10^{-6}
24	4.1×10^{-6}	2.7×10^{-8}
32	1.1×10^{-8}	1.5×10^{-9}

176 that is, as simplify($dF/dx - f$) = 0 rather than by numerical agreement.
 177 (ii) In 40- to 60-digit arithmetic, (7) agrees with a brute-force double integral
 178 of the constant-parameter density over (v, w) to 8.7×10^{-32} , and (10) agrees
 179 with quadrature of (1) to between 10^{-35} and 10^{-48} at $t/a^2 = 0.9, 1.5$, and 4.7 .
 180 (iii) On 400 random parameter sets with $a \in \{0.6, 1, 1.5, 2.5\}$, $\nu \in [-3, 4]$,
 181 $\eta \in \{0.5, 1, 2\}$, $w_2 - w_1 \in \{0.05, 0.2, 0.4\}$, and $t \in [0.05, 3]$, spanning densi-
 182 ties from 5×10^{-20} to 1.6 , the switched representation agrees with WienR
 183 (Hartmann and Klauer, 2021) called at its tightest tolerance (10^{-12}) to a
 184 maximum absolute difference of 1.2×10^{-14} ; rtdists (Singmann et al., 2022),
 185 whose diffusion code is that of fast-dm, converges onto the same values as
 186 its precision setting is raised (7.6×10^{-4} , 1.1×10^{-5} , 2.5×10^{-8} at precision
 187 3, 5, 8), as does the HDDM likelihood (Wiecki et al., 2013) as its integra-
 188 tion tolerance is tightened. On 200 seven-parameter sets, (14) agrees with
 189 WienR to 2.5×10^{-9} , which is WienR’s own reported cubature error. (iv) An
 190 Euler–Maruyama simulation of the process with 400 000 trials, step 5×10^{-5} ,
 191 and the Broadie–Glasserman–Kou barrier correction gives a lower-barrier re-
 192 sponse time histogram in 30 bins of 100 ms whose χ^2 against the binned
 193 density (7) is 21.6, with all standardised residuals below 1.9. (v) The deriva-
 194 tion of (7), including the antiderivative, the coefficient table, the exchange of
 195 summation and integration, and the Faddeeva integral (9), is formalised and
 196 machine-checked in Lean 4 with Mathlib (de Moura and Ullrich, 2021; The
 197 mathlib Community, 2020); the development contains no unproved state-
 198 ment and relies on no axioms beyond Lean’s standard ones. The starting
 199 representations (1) and (8) are taken as definitions there, as they are taken
 200 from the literature here.

201 7.2. Efficiency

202 Table 2 compares the cost of (7) and (14), compiled from a direct tran-
 203 scription of the formulae, with the three packages in common use: WienR,
 204 which evaluates the drift-integrated series by adaptive cubature over w ; rtdists,
 205 whose diffusion code is that of fast-dm and integrates over w and the
 206 non-decision window by adaptive Simpson’s rule; and the HDDM likelihood,
 207 which does the same with the Navarro–Fuss series. All timings are single-
 208 threaded on one machine,² for 10 000 response times at one parameter set,
 209 which is the situation in a likelihood evaluation.

210 The error columns need a reference scale. Blurton et al. (2017) recom-
 211 mend a truncation tolerance of 1.5×10^{-8} , the square root of single-precision
 212 machine epsilon, and Gondan et al. (2014) illustrate their term counts at
 213 10^{-3} . Tuerlinckx (2004, Table 1) reports differences of 2×10^{-7} to 3×10^{-4}
 214 between his quadrature-based algorithm and the classical one. Among cur-
 215 rent packages, rtdists (fast-dm) defaults to a precision of 3, documented as
 216 roughly the number of accurate decimals; the HDDM likelihood truncates
 217 its series at 10^{-4} and integrates over the starting point and the non-decision
 218 window by adaptive Simpson’s rule with tolerance 10^{-8} and recursion depth
 219 10; the Stan implementation of Henrich et al. (2024) fixes the density tol-
 220 erance at 10^{-6} and defaults to 10^{-4} for the derivatives; WienR defaults to
 221 10^{-12} . The closed form is exact to double precision, and its seven-parameter
 222 quadrature at 1.5×10^{-9} is below every one of these tolerances.

223 Two features of Table 2 are worth stating. First, the closed form is the
 224 cheapest of the four at every entry and the only one that is exact for six
 225 parameters. rtdists costs 1.3 times as much for six parameters and twice as
 226 much for seven, and deviates from the density by 0.5% and 0.9%. HDDM
 227 costs 1.8 and 5 times as much and agrees to 4×10^{-6} and 1.5×10^{-7} ; its
 228 series tolerance of 10^{-4} leaves a floor near 10^{-7} that tightening its integration
 229 settings does not remove. WienR reaches 10^{-12} at 6 and 110 times the cost.
 230 Because the cost of the closed form is fixed whereas that of the integrators
 231 is set by the accuracy asked of them, these ratios grow with the tolerance.
 232 Second, the gradient comes almost free: all five partial derivatives of (7) share
 233 the exponential factors of the density itself, and an uncompiled vectorised

²Apple M1 Pro; R 4.5.3 with WienR 0.3-15 and rtdists 0.11-6; Python 3.11 with NumPy 1.26.4, SciPy 1.13.1, Cython 3.1.4, and hddm-wfpt 0.1.4, the HDDM likelihood distributed as a standalone package.

Table 2: Cost per evaluation in microseconds, single thread, 10 000 response times at $a = 1.2$, $\nu = 1$, $w = 0.5$, $\eta = 1$, $w_2 - w_1 = 0.2$, $s_t = 0.2$. Each package at the settings its fitting routine uses: WienR at tolerance 10^{-12} , rtdists at precision 3, HDDM at series tolerance 10^{-4} with Simpson tolerance 10^{-8} and recursion depth 10. Error is the maximum relative deviation from (7) or (14) over the parameter sets of Section 7 (densities above 10^{-3}); for the closed form the seven-parameter entry is the quadrature error of (14) against adaptive integration. [†]Uncompiled vectorised implementation.

	Six parameters		Seven parameters (+ s_t)	
	Error	μs	Error	μs
Closed form	–	0.33	1.5×10^{-9}	9.2
WienR	2.5×10^{-12}	2.1	1.5×10^{-9}	1040
rtdists	4.6×10^{-3}	0.43	9.2×10^{-3}	18
HDDM	3.9×10^{-6}	0.60	1.5×10^{-7}	46
Closed form + 5 gradients [†]	–	1.7		
WienR, 5 gradients	10^{-13}	15.7		

234 implementation returns the density together with all five derivatives in $1.7 \mu s$,
235 whereas WienR requires one further cubature per derivative. For reference,
236 WienR’s own closed form for drift variability alone (Blurton et al., 2017)
237 evaluates in $0.26 \mu s$ on the same machine, so the closed form with starting-
238 point variability costs 1.3 times the drift-only density.

239 8. Discussion

240 We have shown that the first-passage time density of the diffusion model
241 with normally distributed drift and uniformly distributed starting point has a
242 closed form, and that the reason it was not found earlier is the choice of series
243 representation: the small-time series carries a Gaussian factor in the starting
244 point that the large-time series lacks, and that factor is exactly what gives
245 the integral a closed form. The result completes the analytic treatment of
246 the drift and starting-point integrals of the seven-parameter model and leaves
247 only the non-decision-time integral, which is a one-dimensional quadrature
248 of a smooth function on a short interval and for which we have given a fixed
249 32-node rule accurate to 10^{-9} .

250 For practice the consequences are threefold. The density is exact up to
251 a truncation error that is controlled by (11); it is evaluated at a fixed cost
252 that is below that of each existing package at the settings the package fits
253 with; and its partial derivatives are closed form and nearly free, which is

what hierarchical Bayesian and Hamiltonian Monte Carlo implementations (Wiecki et al., 2013; Henrich et al., 2024) need. The result is directly usable in any of the existing packages, which already contain the drift-integrated density (1) as a building block and need only replace the quadrature over w by (6).

The corresponding cumulative distribution function with both forms of variability does not follow by the same route. Starting from the drift-integrated distribution function of Blurton et al. (2017), half of the series terms have a coefficient of w^2 that is exactly zero after the drift integral and close through the integral $\int e^{cw} \Phi(\alpha w + \beta) dw$, which has a closed form; the other half carry the coefficient $+2a^2\eta^2$, a growing Gaussian times Φ , and reduce to an integral of Owen- T type that has no closed form in terms of $\exp$, Φ , or Owen's T . For maximum likelihood and Bayesian estimation the density suffices, and we leave the distribution function open.

Acknowledgments

Appendix A. The antiderivative

Let $A > 0$, $m = B/A$, and

$$F(w) := (p + qm) \sqrt{\frac{2\pi}{A}} e^{C+B^2/2A} \Phi(\sqrt{A}(w - m)) - \frac{q}{A} e^{C+Bw-\frac{1}{2}Aw^2}.$$

Using $\Phi' = \phi$ and $-\frac{1}{2}A(w - m)^2 + C + B^2/2A = C + Bw - \frac{1}{2}Aw^2$,

$$\begin{aligned} F'(w) &= (p + qm) \sqrt{2\pi} e^{C+B^2/2A} \phi(\sqrt{A}(w - m)) - \frac{q}{A} (B - Aw) e^{C+Bw-\frac{1}{2}Aw^2} \\ &= [(p + qm) - q(m - w)] e^{C+Bw-\frac{1}{2}Aw^2} = (p + qw) e^{C+Bw-\frac{1}{2}Aw^2}, \end{aligned}$$

which is the integrand of (6). The definite integral (6) is $F(w_2) - F(w_1)$. The same computation is carried out symbolically in the supplementary code and formally in the Lean development.

Appendix B. Term bounds

For $w \in [0, 1]$, $ja \leq r_j \leq (j + 1)a$, and the prefactor of (1) is bounded by $\exp[(2|\nu|a + \eta^2 a^2)/2S]$. Hence the j -th term of the series in (1), and therefore $|I_j|/(w_2 - w_1)$ in (7), is at most $(j + 1)a \exp(-j^2 a^2/2t)$ times a

constant independent of j and w . Since $\sum_j (j+1) \exp(-j^2 c)$ converges for every $c > 0$, the series converges absolutely and uniformly in w , which justifies the exchange of summation and integration and gives the truncation rule for J in (11). For the large-time series (10), $|w(z)| \leq 1$ in the upper half plane and $|e^{\kappa w^2 + \mu_k w}| \leq e^{\kappa + |\lambda|}$ on $[0, 1]$, so the k -th term is at most a constant times $k \exp(-k^2 \pi^2 t / 2a^2)$, which gives the rule for K .

Appendix C. Supplementary material

The supplementary material contains (i) a vectorised implementation of (7), (10), (14), and all partial derivatives in Python, together with a compiled kernel used for Table 2; (ii) the symbolic verification script; (iii) the scripts and reference values for the comparisons with WienR, rtdists, and HDDM, and the simulation; and (iv) the Lean 4 formalisation.

References

- Abramowitz, M., Stegun, I.A., 1964. Handbook of Mathematical Functions. National Bureau of Standards, Washington, DC.
- Blurton, S.P., Kesselmeier, M., Gondan, M., 2017. The first-passage time distribution for the diffusion model with variable drift. *Journal of Mathematical Psychology* 76, 7–12.
- Gondan, M., Blurton, S.P., Kesselmeier, M., 2014. Even faster and even more accurate first-passage time densities and distributions for the Wiener diffusion model. *Journal of Mathematical Psychology* 60, 20–22.
- Hartmann, R., Klauer, K.C., 2021. Partial derivatives for the first-passage time distribution in Wiener diffusion models. *Journal of Mathematical Psychology* 103, 102550.
- Henrich, F., Hartmann, R., Pratz, V., Voss, A., Klauer, K.C., 2024. The seven-parameter diffusion model: An implementation in Stan for Bayesian analyses. *Behavior Research Methods* 56, 3102–3116. doi:10.3758/s13428-023-02179-1.
- Horrocks, J., Thompson, M.E., 2004. Modeling event times with multiple outcomes using the Wiener process with drift. *Lifetime Data Analysis* 10, 29–49.

- 310 de Moura, L., Ullrich, S., 2021. The Lean 4 theorem prover and programming
311 language, in: Automated Deduction – CADE 28, Springer. pp. 625–635.
- 312 Navarro, D.J., Fuss, I.G., 2009. Fast and accurate calculations for first-
313 passage times in Wiener diffusion models. *Journal of Mathematical Psy-*
314 *chology* 53, 222–230.
- 315 Ratcliff, R., 1978. A theory of memory retrieval. *Psychological Review* 85,
316 59–108.
- 317 Ratcliff, R., Rouder, J.N., 1998. Modeling response times for two-choice
318 decisions. *Psychological Science* 9, 347–356.
- 319 Ratcliff, R., Tuerlinckx, F., 2002. Estimating parameters of the diffusion
320 model: Approaches to dealing with contaminant reaction times and pa-
321 rameter variability. *Psychonomic Bulletin & Review* 9, 438–481.
- 322 Singmann, H., Brown, S., Gretton, M., Heathcote, A., 2022. rtdists: Re-
323 sponse time distributions. R package version 0.11-6.
- 324 The mathlib Community, 2020. The Lean mathematical library, in: Pro-
325 ceedings of the 9th ACM SIGPLAN International Conference on Certified
326 Programs and Proofs, pp. 367–381.
- 327 Tuerlinckx, F., 2004. The efficient computation of the cumulative distribution
328 and probability density functions in the diffusion model. *Behavior Research*
329 *Methods, Instruments, & Computers* 36, 702–716.
- 330 Vandekerckhove, J., Tuerlinckx, F., 2008. Diffusion model analysis with
331 MATLAB: A DMAT primer. *Behavior Research Methods* 40, 61–72.
- 332 Voss, A., Voss, J., 2007. Fast-dm: A free program for efficient diffusion model
333 analysis. *Behavior Research Methods* 39, 767–775.
- 334 Wiecki, T.V., Sofer, I., Frank, M.J., 2013. HDDM: Hierarchical Bayesian
335 estimation of the drift-diffusion model in Python. *Frontiers in Neuroinfor-*
336 *matics* 7, 14.
- 337 Zaghloul, M.R., Ali, A.N., 2011. Algorithm 916: Computing the Faddeyeva
338 and Voigt functions. *ACM Transactions on Mathematical Software* 38, 15.